CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (ME/EE /EC/CL) **SEM - III** **University Examination, 2010-11**

MA – 202 Engineering Mathematics-III

Date: 16.05.11, Monday **Time: 10:00 a.m to 01:00 p.m** **Maximum Marks: 70**

Instructions: (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book; don't write anything on the question paper.

SECTION-I

Q-1 (a) Show that $x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}(-1)^n \frac{\cos nx}{n^2}$ in the interval $-\pi \le x \le \pi$. **[05]**
Hence deduce that $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - ... = \frac{\pi^2}{12}$.

(b) Find the inverse Laplace transform of $\frac{s+7}{s^2+2s+2}$. **[03]**

(c) (i) Define Laplace transforms (ii) state convolution theorem (iii) $L\left[e^{at}\sinh bt\right] =$ __________. **[03]**

Q-2 (a) Find a Fourier series to represent $x - x^2$ from $x = -\pi$ to $x = \pi$. **[05]**

(b) Solve $\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + x = e^t$ with $x(0) = 2$, $x'(0) = -1$ using Laplace transforms. **[05]**

(c) Find the Laplace transforms of $e^{-3t}(\cos 4t + 3\sin 4t)$. **[02]**

OR

Q-2 (a) Find the Laplace transforms of the function $f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & t > \pi \end{cases}$ **[05]**

(b) Find the Fourier series to represent $f(x) = x^2 - 2$ when $-2 \le x \le 2$. **[05]**

(c) If $f(x) = x\cos x$ in $(-\pi, \pi)$ then the value of $a_n =$ ______. **[02]**

Q-3 (a) Solve the cubic equation by Cardon's method: $x^3 - 18x - 35 = 0$. **[05]**

(b) Solve the equation $x^4 + 2x^3 - 7x^2 - 8x + 12 = 0$ by Ferrari's method. **[05]**

(c) Find the Laplace transforms of $\sin(at + b)$. **[02]**

OR

Q-3 (a) Solve the equation $x^3 - 15x - 126 = 0$, by Cardon's method. **[05]**

(b) Solve the equation $x^4 + 12x - 5 = 0$ by Ferrari's method. **[05]**

(c) Find the Laplace transforms of $\sin 2t \cos 3t$. **[02]**

SECTION-II

Q-4 (a) Find the angle between the tangents to the curve $x = t^2, y = 2t, z = -t^3$ at the points $t = 1$ and $t = -1$. **[04]**

(b) A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2\cos 3t$ and $z = 2\sin 3t$, where $'t'$ is the time. **[04]**

1) Determine its velocity and acceleration at any time.
2) Find the magnitudes of the velocity and acceleration at $t = 0$.

(c) (i) State Green's theorem (ii) State Divergence Theorem (iii) Define Directional Derivatives **[03]**

Q-5 (a) Using Green's theorem, evaluate $\oint_C \left(3x^2 - 8y^2\right)dx + \left(4y - 6xy\right)dy$, where C is the boundary of the region bounded by $y^2 = x$ and $y = x^2$. **[05]**

(b) Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at the point $(2,-1,2)$ in the direction $2\hat{i} + 3\hat{j} + 6\hat{k}$. **[04]**

(c) A fluid motion is given by $\vec{V} = (y+z)\hat{i} + (z+x)\hat{j} + (x+y)\hat{k}$. Is this motion irrotational? **[03]**

OR

Q-5 (a) Using divergence theorem, evaluate $\int_S \vec{F} \cdot d\vec{S}$, where $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ and S is the surface bounding the region $x^2 + y^2 = 4, z = 0$ and $z = 3$. **[05]**

(b) Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at the point $(2,-1,2)$ towards the point $(1,1,-1)$. **[04]**

(c) Find $curl\,\vec{F}$, where $\vec{F} = grad\left(x^3 + y^3 + z^3 - 3xyz\right)$ **[03]**

Q-6 (a) Using Stoke's theorem, evaluate $\int_C \left[(x+y)dx + (2x-z)dy + (y+z)dz\right]$, where C is the boundary of the triangle with vertices $(2,0,0), (0,3,0)$ and $(0,0,6)$. **[05]**

(b) A body executes damped forced vibrations given by the equation, **[05]**

$\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 64x = 128\cos 8t$. Solve the equation, when $x = \frac{1}{3}, \frac{dx}{dt} = 0$ at $t = 0$.

(c) Find the value of $'b'$ for which the vector field $\vec{F} = (2x+y)\hat{i} + (3x-2z)\hat{j} + (x+bz)\hat{k}$ is solenoidal. **[02]**

OR

Q-6 (a) If $\nabla\phi = y^2\hat{i} + \left(2xy + z^3\right)\hat{j} + 3yz^2\hat{k}$, determine ϕ. **[05]**

(b) Show that the frequency of free vibrations in a closed electrical circuit with inductance L and capacity C in series is $\frac{30}{\pi\sqrt{LC}}$ cycles/minute. **[05]**

(c) Find the value of $'a'$ for which the vector $\vec{V} = (y+2)\hat{i} + ax\,\hat{j} + 2z\,\hat{k}$ is irrotational. **[02]**
